

THE TWO BOOKS

two kinds of ground · B adds them

FAST PROVISION A_f

$b_f \cdot A_f$ · high yield, quick
small standing value



conversion $u_c \cdot A_c \downarrow A_f \uparrow$

ECOLOGICAL CAPITAL A_c

$Y_c \cdot A_c$, big standing value
 $b_c + b_{GG}(q)$



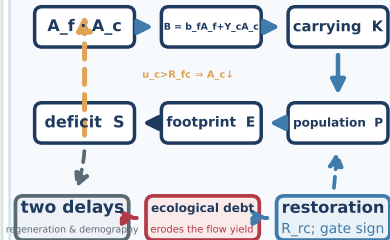
BIOCAPACITY B

$= b_f A_f + Y_c A_c$
area conserved; capacity a state

composition condition: $b_f - b_{c,eff} > 0$
 $\Rightarrow$ converting A_c into A_f raises B

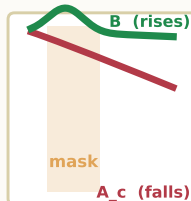
THE COUPLING

loop, deficit switch, debt, delays



THE COMPOSITION ILLUSION

B up while A_c down



B $\uparrow$ while $A_c \downarrow$
the composition illusion

demand $> E_{ceil} \approx 1.14$
 $A_c \rightarrow$ typed floor (floor collision, not a fold)

no critical slowing
leader stays $\approx -0.054 \text{ yr}^{-1}$

composition step
NOT identifiable from B alone

Recovery needs a gated restoration flow;
the collapse trigger is demand, not a delay.

measured 1961-2022: cropland 2.85 $\times$, built-up 8.63 $\times$
aggregate +24.5% · non-cropland $\approx$ flat (1.01 $\times$)

How Aggregation Can Conceal Composition: Aggregate Biocapacity and the Identifiability of Modelled Ecological-Capital Drawdown